



Habib University

EE-437 Renewable energy systems – Fall 2025

Time = 40 minutes

Final Exam Part 3

Max Points: 20

Question 14 [2 points]: A coal power plant with rated capacity of 1500MW with capex at \$2,500/kW is expected to last for 20 years. Its fixed O&M is \$25/kW per year, and fuel cost is \$40/MWh. Its capacity is 85%. Calculate its LCOE in terms of \$/kWh (Hint: The denominator will be total energy produced). Simplify the formula as much as possible, before needing to expand the summation.

$$\frac{\text{Energy}}{\text{year}} = 1500 * 10^3 * 0.85 * 8760 = 1.1169 * 10^{10} \text{ kwh/year}$$

$$LCOE = \frac{[(1500 * 10^3) * 2500] + \sum_{t=1}^{20} \frac{\{(1500 * 10^3 * 25) + (1.1169 * 10^{10} * 40 * 10^{-3})\}}{(1+i)^t}}{1.1169 * 10^{10} * 20}$$

Question 15 [3 points]: Choose between two sites for wind power. Site A has constant winds of 8m/s for every hour of the year. Site B has 12m/s winds for 6 months of the year and 3m/s winds for other half of the year. Which site will have more energy output? What advantage does site A have over site B and vice versa? Give justifications.

$$\text{Energy}_A = 0.5 * c_p * A * \rho * 8760 * v^3 = k * 8760 * v^3 = k * 4,485,120$$

$$\begin{aligned} \text{Energy}_B &= 0.5 * c_p * A * \rho * \frac{8760}{2} * v_1^3 + 0.5 * c_p * A * \rho * \frac{8760}{2} * v_2^3 \\ &= k * 7,568,640 + k * 118,260 = k * 7,686,900 \end{aligned}$$

Site B will produce more energy. Site A is more steady/predictable.

Question 16[3 points]: A house in Karachi requires 30 kWh/day of electricity. You are tasked with sizing a PV array to meet this demand. Find the minimum number of panels needed to meet this requirement.

Peak Sun Hours (PSH): 5.5 h/day, PV module rated power: 500 Wp (at STC: 20°C, 1000 W/m²), Module efficiency at STC: 18%, Temperature coefficient of efficiency: -0.45%/°C, Average cell operating temperature: 50 °C, System performance ratio (wiring, inverter, dust losses, etc.): 0.75

$$\begin{aligned} \text{Temperature loss} &= -0.45 \% \times 30^\circ\text{C} = -13.5\% \\ P_{eff} &= 500 \text{ W} \times (1 - 0.135) = 500 \times 0.865 = 432.5 \text{ W} \\ P_{use} &= 364 \text{ W} \times 0.75 = 273 \text{ W} \end{aligned}$$

$$E_{panel} = P_{use} \times PSH = 532.5 \text{ W} \times 5.5 \text{ h} = 2378.75 \text{ Wh} = \frac{2.37875 \text{ kWh}}{\text{day}}$$

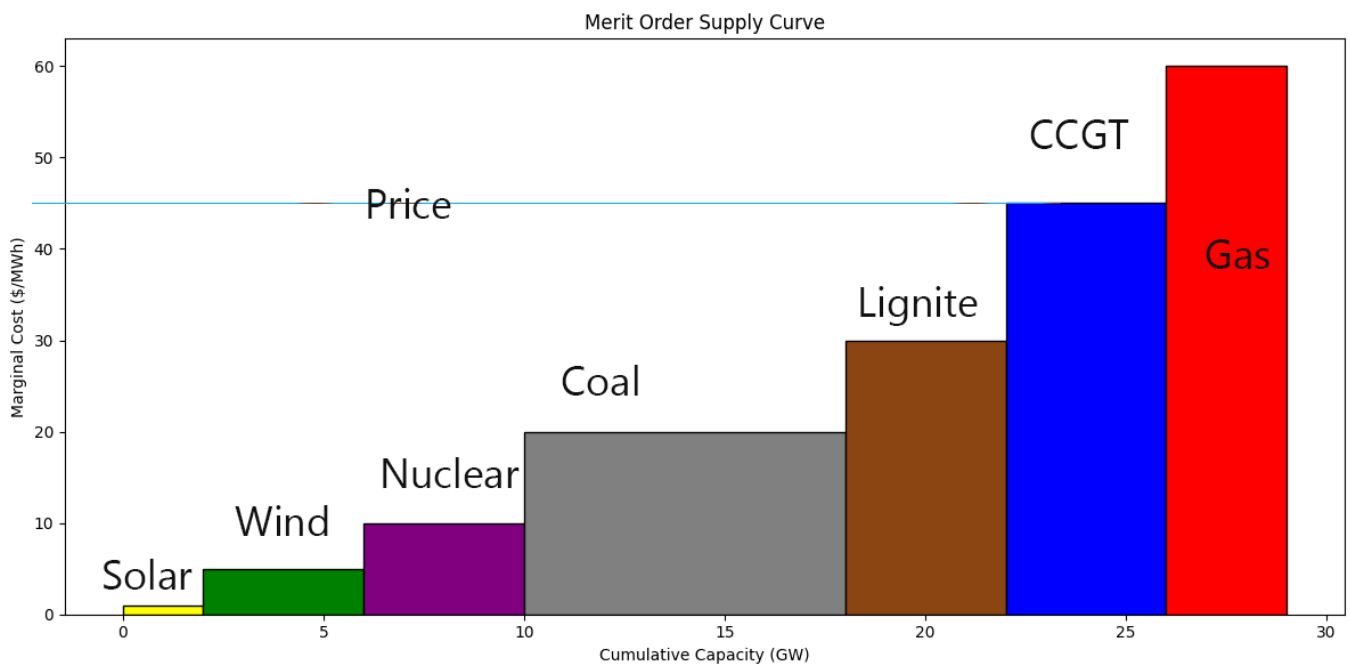
$$N = \frac{30 \text{ kWh}}{2.37875 \text{ kWh}} \approx 12.6 \Rightarrow \mathbf{13 \text{ panels (must be integer)}}.$$

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Question 17 [5 points]: For a given hour, you are given the following data:

Plant	Marginal cost (\$/MWh)	Capacity (GW)
Coal	20	8
CCGT	45	4
Solar	0	2
Lignite	30	4
Nuclear	10	4
Gas	60	3
Wind	5	4

Build the merit order for this and find the price of electricity for this hour if demand is 25GW. How much profit will the nuclear and CCGT plants earn for this hour?



$$Profit_{nuclear} = (45 - 10) * 4000 = \$140,000$$

$$Profit_{CCGT} = (45 - 45) * 4000 = \$0$$

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Question 18 [7 points]: Based on the load and solar data and data for plants, show when each plant is used, with battery and without battery.

Calculate the total savings with storage.

Coal power plants: 4 GW Installed capacity, 10 \$/MWh Operational costs.

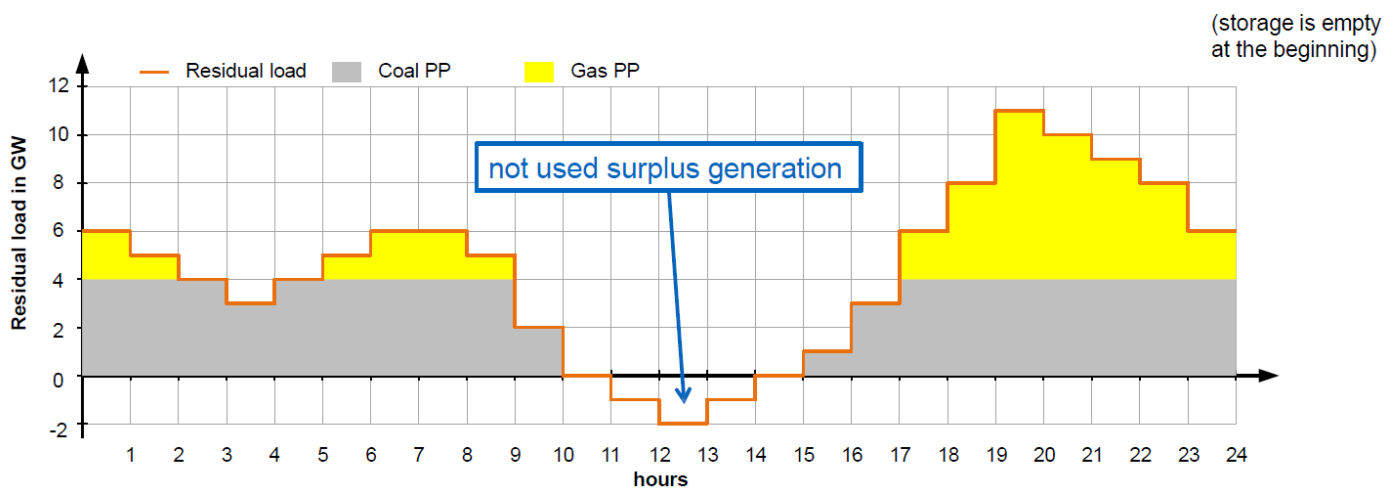
Gas power plants: 8 GW Installed capacity, 25 \$/MWh Operational costs:

Storage unit: 2 GW Installed power capacity (charging/discharging),

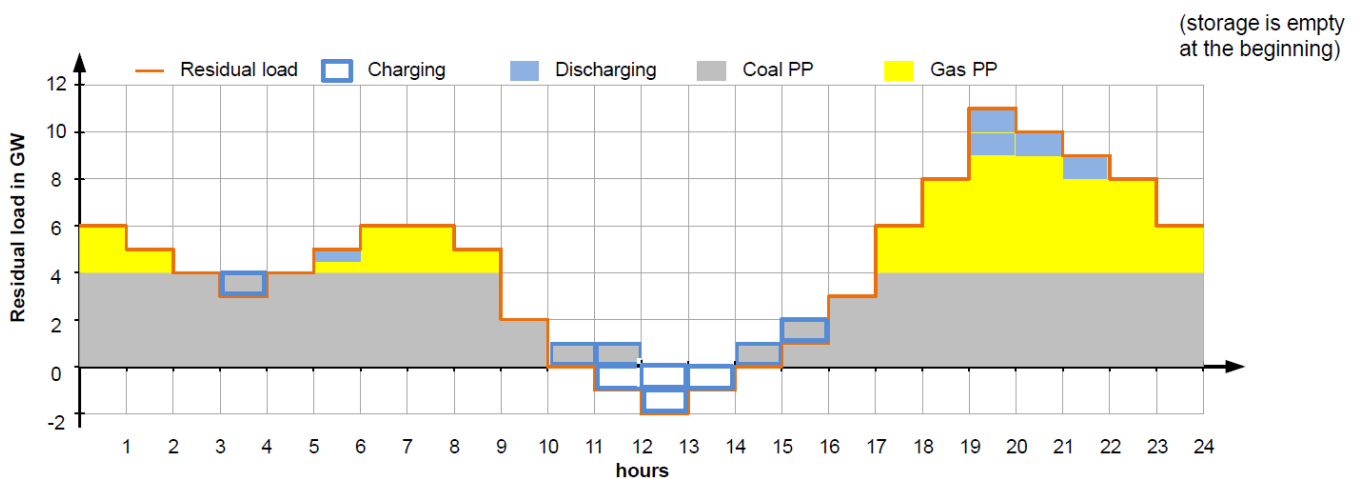
4 GWh Installed storage capacity (discharging), 50% Efficiency

Ans:

Without battery:



With battery:



Extra production of coal = 5GWh (cost = 5000×10)

Reduced production of gas = 4.5GWh (cost = 4500×25)

Savings with battery = \$62,500